Surround Sound V2: Real-Time Environmental, Sound Event, and Speech Recognition from Microphone Audio

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Abstract

Real-time auditory scene understanding is a major challenge in machine learning, particularly outside the domains of speech recognition and music information retrieval. This project, *Surround Sound V2*, extends the original system with a dedicated speech classification head and live ASR transcription, producing a unified pipeline capable of interpreting not only what is happening in an acoustic environment, but also what is being said within it. Using large-scale datasets including AudioSet, FSD50K, CHiME-6, and Mozilla Common Voice, the system trains three convolutional neural network (CNN) classifiers on log–mel spectrogram representations: an eight-class environment classifier, a 208-label multi-label sound event detector, and a three-class speech detector. A unified preprocessing pipeline standardizes all audio to 16 kHz mono, normalized ten-second segments, and consistent log–mel features.

Training incorporates SE channel attention, residual skip connections, focal loss, SpecAugment augmentation, and random hyperparameter search. Results show substantial improvements over V1 across all tasks: environment classification rises from macro-F1 ≈ 0.50 to 0.75, sound event detection rises from macro-F1 ≈ 0.39 to 0.55 despite tripling the label space to 208 classes, and the new speech classifier achieves 96.5% accuracy. A supplemental real-time demo integrates all three classifiers with Whisper large-v3 transcription and a GPT-4.1 Mini scene summarizer, showcasing the system's practical potential for live acoustic awareness. This paper details the datasets, preprocessing, training pipeline, evaluation results, and future directions for unifying environment, event, and speech recognition into a cohesive auditory framework.

Keywords

Audio Classification, Acoustic Scene Recognition, Sound Event Detection, Speech Detection, Automatic Speech Recognition, Log-Mel Spectrograms, Real-Time Systems, Whisper

1 Introduction

Machine learning for audio understanding has advanced rapidly in domains such as automatic speech recognition, music tagging, and medical acoustics. Yet general-purpose auditory scene awareness—the ability for a system to interpret what is happening around it based solely on sound—remains comparatively underexplored. Unlike speech or music, real-world acoustic environments contain overlapping sources, variable noise conditions, and highly imbalanced events, making the problem difficult to organize, process, and evaluate.

Surround Sound V2 extends the original system by addressing two limitations that remained after V1: the absence of any speech-aware component, and comparatively modest classification performance attributable to limited dataset coverage, architectural simplicity, and a narrow event taxonomy. V2 adds a dedicated three-class speech classifier, integrates Whisper large-v3 for live ASR transcription, expands the sound event taxonomy from 62 to 208 labels, and introduces several modern deep learning improvements across all three models. Together, these additions enable the system to interpret an acoustic scene in broad contextual terms (e.g., *indoor*, *crowd*, *vehicle*), fine-grained auditory details (e.g., *human laugh*, *glass clink*, *wind*, *music*), and spoken language—reflecting more fully how a person would describe what they hear.

1.1 DCASE and Related Work

The Detection and Classification of Acoustic Scenes and Events (DCASE) Challenge series [9] has played a central role in advancing acoustic machine listening. DCASE tasks are typically divided into two categories: *Acoustic Scene Classification* (ASC), which assigns a single global label to an audio segment, and *Sound Event Detection* (SED), which identifies one or more overlapping events. Overall, DCASE provides carefully curated benchmarks that demonstrate what is achievable under controlled conditions, but they are not designed as end-to-end, real-time scene understanding systems.

Surround Sound V2 diverges from these paradigms in four ways:

1. **Multi-head design.** Rather than framing the problem as ASC *or* SED, this work trains three complementary classifiers that operate simultaneously, enabling richer interpretation of a single audio segment.
2. **Speech awareness.** A dedicated speech detector and Whisper large-v3 transcription extend the pipeline beyond acoustic scene understanding into spoken content, a capability absent from standard DCASE tasks.
3. **Unified preprocessed datasets.** AudioSet, FSD50K, CHiME-6, and Mozilla Common Voice provide a diverse mix of real-world audio curated and labeled from online and naturalistic recordings [2, 3]. Light preprocessing harmonizes all sources into compatible training inputs.
4. **Real-time, user-facing inference.** Unlike DCASE benchmarks, which evaluate offline performance on curated datasets, this project targets real-time deployment, including microphone input, spectrogram computation, and an optional natural-language summarization stage powered by an LLM.

1.2 Limitations of Existing Approaches

Existing ASC and SED systems face several challenges that motivate an integrated approach:

- **Dataset noise and availability.** AudioSet clips are sourced from YouTube, meaning many recordings become unavailable over time, resulting in incomplete or noisy training sets [3].
- **Severe class imbalance.** Many sound events occur extremely rarely, while others dominate the dataset. This imbalance greatly affects supervised learning.
- **Lack of real-time considerations.** Most models are optimized for batch inference rather than streaming inputs.
- **No speech integration.** ASC and SED systems treat speech as just another event label rather than a distinct modality warranting dedicated detection and transcription.

These limitations motivate building a flexible, modular system rather than a single unified classifier.

1.3 Contributions

This project makes the following contributions:

- A complete dataset acquisition pipeline for AudioSet environment clips, FSD50K event audio, CHiME-6 multi-speaker recordings, and Mozilla Common Voice speech clips, including robust downloading, metadata generation, and label handling.
- A unified preprocessing framework that converts all audio into normalized, fixed-length, log–mel spectrogram tensors suitable for CNN-based learning.
- Three lightweight but effective convolutional neural network models: an eight-class environment classifier, a 208-label multi-label sound event detector, and a three-class speech detector.
- Integration of Whisper large-v3 for real-time ASR transcription, conditioned on speech detection output.
- Architectural improvements applied across all models: SE channel attention, residual skip connections, focal loss, SpecAugment augmentation, and random hyperparameter search.
- A real-time inference and visualization demo that integrates all three classifiers with Whisper transcription and GPT-4.1 Mini scene summarization.
- A detailed analysis of performance, failure modes, and limitations across all three tasks, with direct comparisons to V1.

1.4 Paper Outline

The remainder of this paper is organized as follows. Section 2 reviews relevant literature in acoustic scene recognition, sound event detection, and speech classification. Section 3 describes the datasets and label taxonomies used. Section 4 outlines the preprocessing and feature extraction pipeline. Section 5 details the model architectures and training procedures. Section 6 reports experimental results for all three classifiers with direct V1 comparisons. Section 7 presents the supplemental real-time demo system. Finally, Section 8 concludes with future directions.

2 Related Work

Acoustic Scene Classification (ASC) assigns a single global label (e.g., *park*, *bus*, *office*) to an audio segment, achieving strong performance on curated benchmarks such as the DCASE challenges [9]. However, ASC models are typically evaluated offline and do not capture the fine-grained acoustic events embedded within a scene. Sound Event Detection (SED) complements ASC by identifying one or more overlapping events, often using multi-label CNN architectures with sigmoid outputs and threshold tuning, similar to the events classifier used in this project.

Large-scale datasets such as AudioSet [3] and FSD50K [2] have accelerated progress in ASC and SED but introduce challenges due to missing clips, corrupted downloads, and long-tailed label distributions. These limitations directly affect reproducibility and motivate robust preprocessing and metadata handling, as implemented in Surround Sound V2.

Spectrogram-based CNN architectures similar to those used here have been widely adopted for large-scale audio tagging and environmental sound classification [5, 13]. The Squeeze-and-Excitation mechanism [6] has shown consistent gains through learned channel-wise recalibration and is applied in V2 to all three classification heads. Residual skip connections [4] address gradient flow in deeper networks and are similarly applied across models.

For speech classification, CHiME-6 [14] provides far-field multi-speaker home recordings, offering ecologically valid conditions absent from studio-quality corpora. Whisper large-v3 [12] serves as the ASR backbone; trained on 680K hours of multilingual audio, it is the strongest publicly available open-weight ASR model. Focal loss [7], originally proposed for object detection under extreme class imbalance, is applied here to both the multi-class environment classifier and the multi-label event detector, motivated by the severe long-tail distributions in AudioSet and FSD50K.

Overall, this project builds on established spectrogram-based CNN pipelines while addressing practical considerations that are less emphasized in prior ASC/SED research, including dataset volatility, class imbalance, speech integration, and real-time inference requirements.

3 Datasets

This project uses four large-scale, publicly available datasets to train the environment, sound event, and speech classifiers. Although all sources provide substantial coverage of real-world audio, they differ in taxonomy, labeling methodology, and reliability, requiring additional harmonization and preprocessing.

3.1 AudioSet Environment Subset

AudioSet consists of more than two million 10-second audio clips sourced from YouTube and annotated with over 500 ontology-based labels [3]. For this project, a focused subset of eight high-level environment categories was selected: *animal*, *crowd*, *indoor*, *other_ambient*, *outdoor_nature*, *vehicle_ground*, *vehicle_water*, and *water*. These classes were chosen based on the broad semantic groupings present in the AudioSet ontology, their suitability as contextual "scene" descriptors, and their alignment with the project's goal of real-time environmental awareness.

In V2, VGGish embeddings were downloaded in place of raw WAV audio where available, reducing storage requirements. The download pipeline was improved to recover additional clips, yielding approximately 15K+ balanced and 88K+ unbalanced samples — substantially larger than the 11,207 samples available in V1.

3.2 FSD50K Event Dataset

For fine-grained sound event detection, this work uses FSD50K, an open dataset of human-labeled sound events across 200+ categories [2]. Unlike AudioSet, FSD50K audio files are directly hosted on Zenodo and are consistently accessible.

In V2, the event taxonomy is expanded from the 62-class canonical taxonomy used in V1 to the full 208-class label space based on the AudioSet ontology. The V1 taxonomy merged semantically related labels into coarse groups, collapsing acoustically distinct events and limiting per-class discriminability. Preserving the full hierarchy at 208 classes is offset by the improved focal loss formulation described in Section 5. All 51,197+ available samples were processed for model training, with clips trimmed from their original 30-second maximum to 10-second windows.

3.3 Speech Dataset: CHiME-6 and Common Voice 25.0

The speech classifier is new in V2 and has no V1 equivalent. Two datasets were combined to produce a 42K-clip training corpus:

- **CHiME-6 (OpenSLR 150) [14].** Far-field multi-speaker home recordings. 17K segmented clips were extracted using 10-second sliding windows, providing realistic overlapping speech conditions absent from studio-quality corpora.
- **Mozilla Common Voice 25.0.** 25K single-speaker read speech clips in English, covering a broad range of speaker demographics. Only 447 clips were available in the *no_speech* class, creating a significant class imbalance discussed further in Section 6.

All speech clips are processed using the same 16 kHz mono log–mel spectrogram pipeline as the environment and event models, enabling the unified preprocessing backbone to serve all three classification heads.

3.4 Dataset Statistics

After filtering and preprocessing, the final datasets used in training consist of:

- **Environment clips:** 15K+ balanced and 88K+ unbalanced samples, 85/15 train/val split.
- **Event clips:** 51K+ samples across 208 classes, 85/15 train/val split.
- **Speech clips:** 42K+ samples (17K CHiME-6, 25K Common Voice), 85/15 train/val split.

0: animal	1: crowd
2: indoor	3: other_ambient
4: outdoor_nature	5: vehicle_ground
6: vehicle_water	7: water

Table 1: Eight-class environment taxonomy used for classification.

0: No Speech	1: Single Speaker	2: Multi Speaker

Table 2: Three-class speech taxonomy used for classification.

Overall, the combined dataset preparation pipeline produces clean, uniformly formatted, and label-consistent log–mel spectrograms suitable for supervised learning across all three classification tasks. The 208-class event taxonomy table is omitted here for brevity; the full label set is available in the project repository.

4 Preprocessing and Feature Extraction

All three classifiers operate on log–mel spectrograms derived from 10-second mono audio segments. A unified preprocessing pipeline standardizes raw audio from all four datasets into consistent feature tensors suitable for supervised learning.

4.1 Waveform Standardization

All clips are converted to 16 kHz mono and normalized to a fixed 10-second duration. Environment audio from AudioSet is preprocessed using `environment_preprocess.py`, which scans both the balanced and unbalanced environment directories, avoids duplicate files, and processes each unique WAV file. Event audio from FSD50K is handled by `events_preprocess.py` based on a manifest that records clip IDs and file paths. Speech audio from CHiME-6 and Common Voice is handled by `speech_preprocess.py`; CHiME-6 recordings are segmented into 10-second sliding windows, and Common Voice clips are zero-padded or trimmed to match. FSD50K clips were additionally trimmed from their original 30-second maximum to 10-second windows prior to spectrogram computation.

For each input file, the pipeline performs:

(1) **Loading and resampling.** Raw audio is loaded with `librosa` [8] at 16 kHz in mono.
(2) **Peak normalization.** The waveform is scaled by its maximum absolute value to avoid clipping and ensure consistent dynamic range.
(3) **Duration fixing.** If the clip is shorter than 10 seconds, zero-padding is applied; if longer, it is center-cropped to exactly 10 seconds.

The resulting standardized waveforms are written back to disk as cleaned WAV files for reproducibility and potential reuse.

4.2 Log–Mel Spectrograms

From each normalized waveform, a log–mel spectrogram is computed using a common configuration across all three tasks:

- Sample rate: 16 kHz
- Number of mel bands: 128
- Hop length: 512 samples
- Maximum frequency: 8 kHz

The mel power spectrogram is converted to decibels using `librosa` [8], with values clipped to a fixed decibel range (approximately $[-80, 0]$) for numerical stability. Features are stored as NumPy arrays of shape (mel, time) under per-task directories. During training, a channel dimension is added so that each example has shape $(1, F, T)$, where F and T represent frequency and time, respectively.

4.3 SpecAugment

V2 applies SpecAugment [10] data augmentation during training for all three models. Frequency masking randomly zeros out contiguous bands along the mel frequency axis; time masking randomly zeros out contiguous frames along the temporal axis. This regularization technique, originally developed for ASR, improves model generalization by preventing over-reliance on specific spectral or temporal regions. Augmentation is applied only during training and disabled at validation and inference time.

4.4 Index Files and Metadata

To connect features with labels and metadata, all pipelines construct index files that summarize usable samples. For the environment head, `01_environment_setup.py` merges metadata from the balanced and unbalanced AudioSet subsets, maps raw label strings to the eight environment categories, and produces a `data_index` file storing the feature path and integer label ID for each clip. `01_events_setup.py` processes FSD50K metadata to create the 208-class taxonomy and multi-label index with per-clip lists of label IDs. `01_speech_setup.py` merges CHiME-6 and Common Voice manifests and maps each clip to one of three speech classes. All index files are saved as both CSV and Parquet and are the only sources consulted by the training scripts. Any clip that fails preprocessing is excluded.

4.5 Consistency Across Tasks

Using a unified preprocessing backbone for all three models provides several benefits:

- **Shared feature space:** All three models learn from spectrograms with identical frequency × time resolution and scaling.
- **Simplified deployment:** The real-time demo computes a single log–mel representation and feeds it to all three heads without redundant computation.
- **Easier experimentation:** Changes to spectrogram parameters or normalization strategies propagate across all models and datasets uniformly.

This standardized preprocessing pipeline forms the foundation for the CNN architectures and training procedures described in Section 5.

5 Models and Training

All three classification heads in the Surround Sound V2 system use convolutional neural networks operating on log–mel spectrogram inputs. V2 introduces two architectural upgrades applied uniformly across all models: Squeeze-and-Excitation (SE) channel attention [6] and residual skip connections [4]. Although the three models share this backbone, they differ in output dimensionality and loss formulation due to the single-label vs. multi-label vs. three-class nature of their respective tasks [5, 13].

5.1 Architectural Upgrades

Squeeze-and-Excitation Channel Attention. After each convolutional block, an SE module performs global average pooling to produce a channel descriptor, passes it through two fully connected layers with ReLU and sigmoid activations respectively, and multiplies the result back as a channel-wise scaling factor. This allows the network to learn which feature channels are most informative for the current input, dynamically suppressing less relevant spectral patterns.

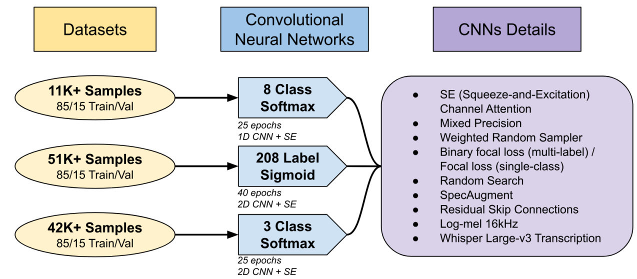

Figure 1: Overview of the environment, event, and speech CNN architectures used in V2.

Residual Skip Connections. Skip connections bypass each convolutional block, adding the block input to its output before activation. This addresses gradient flow in deeper networks and enables the model to learn residual corrections rather than full transformations, improving convergence stability particularly in the event classifier's deeper stack.

Random Hyperparameter Search. Rather than using fixed hyperparameters, V2 applies random search over learning rate, batch size, dropout rate, focal loss gamma, SE reduction ratio, and base channel depth for all three models. The best-performing configuration on the validation set is selected for final training.

5.2 Environment Classifier

The environment classifier predicts one of eight mutually exclusive scene labels. Its V2 architecture extends the V1 four-block CNN with SE attention modules and residual connections. Each block contains:

- a 2D convolution (`Conv2d`) with kernel size 3×3,
- batch normalization,
- ReLU activation,
- 2×2 max pooling,
- an SE channel attention module.

After the convolutional stack, global average pooling reduces spatial dimensions, followed by a fully connected layer that outputs an 8-dimensional logit vector. A final softmax operation produces class probabilities.

The network is trained using focal loss [7] with class weighting to compensate for class imbalance, replacing the cross-entropy loss used in V1. Optimization uses Adam with a base learning rate selected via random search and cosine annealing. Training is performed for 25 epochs using mixed precision (`torch.cuda.amp`) to reduce memory usage.

5.3 Event Classifier

The event classifier predicts the presence or absence of 208 canonical sound events derived from FSD50K metadata — expanded from 62 classes in V1. Because multiple events may co-occur in each clip, the classifier uses a multi-label formulation with sigmoid activations.

The V2 architecture follows a deeper convolutional stack than the environment model to accommodate the greater label complexity, with SE attention and residual connections applied throughout. The model includes:

- three convolutional stages with channel depths from 64 to 256,
- batch normalization and ReLU activations,
- max pooling between stages,
- SE channel attention after each stage,
- global average pooling,
- a 208-dimensional output layer.

A sigmoid function is applied independently to each output dimension. Training uses binary focal loss [7] with logits, replacing the BCEWithLogitsLoss used in V1. Binary focal loss down-weights easy negatives more aggressively than standard BCE and is particularly suited to the long-tail distribution of FSD50K, where dominant classes would otherwise overwhelm rare but acoustically distinct events. Class-dependent weights computed from label frequencies are additionally passed into the loss function. The threshold for event presence is tuned globally on the validation set to optimize macro-F1. Training runs for 40 epochs with Adam and cosine annealing.

5.4 Speech Classifier

The speech classifier is new in V2. It predicts one of three mutually exclusive classes — *no_speech*, *single_speaker*, *multi_speaker* — from a log–mel spectrogram input. The architecture is a 2D CNN with SE attention and residual connections, structurally similar to the environment classifier but with a 3-dimensional softmax output.

The three-class softmax formulation treats speech detection as mutually exclusive: at any given 10-second window, one category dominates. Training runs for 25 epochs using focal loss, with random search over learning rate, batch size, dropout, focal gamma, SE reduction ratio, and base channel depth. The severe imbalance between *no_speech* (447 clips) and the two speech classes (42K+ clips combined) is the primary training challenge and is addressed through focal loss gamma weighting.

When the speech classifier predicts *single_speaker* or *multi_speaker* above a confidence threshold, the audio segment is passed to Whisper large-v3 [12] via the `faster-whisper` library for full ASR transcription. The transcript is incorporated into the GPT-4.1 Mini scene summarization prompt alongside the environment and event predictions.

5.5 Batching, Augmentation, and Sampling

Training uses batch sizes selected via random search in the range 32–128. Spectrograms are stacked into tensors of shape $(B, 1, F, T)$, where F=128 mel bands and T varies with hop length. SpecAugment [10] is applied during training for all three models: frequency masking randomly zeros contiguous mel bands and time masking randomly zeros contiguous temporal frames. This replaces the lightweight augmentation used in V1, which showed limited gains.

For the event model, weighted random sampling addresses extreme class imbalance during batch construction, ensuring rare classes receive adequate representation without oversampling individual clips.

5.6 Training Stability and Convergence

All three models benefit from mixed-precision training and cosine learning-rate schedules, which improve stability over long training runs on a 16 GB VRAM GPU. Early stopping is avoided in favor of full training cycles. Random hyperparameter search selects the best checkpoint per model based on validation macro-F1.

The final environment, event, and speech classifiers reach stable convergence and are summarized in Section 6.

6 Experimental Results

This section summarizes the performance of all three classifiers on held-out validation sets. All results use log–mel spectrogram inputs generated by the unified preprocessing pipeline described in Section 4. We train all models using PyTorch [11] on a laptop with an NVIDIA GeForce RTX 5080 Laptop (16 GB VRAM) and a Intel(R) Core(TM) Ultra 9 285H (2.90 GHz) CPU. All datasets are split 85/15 into training and validation sets as described in Section 3.

6.1 Environment Classification Results

Table 3 presents the macro- and weighted-average metrics for the V1 and V2 environment classifiers. V2 achieves substantial improvements across all aggregate metrics, with macro-F1 rising from 0.4985 to 0.7537.

Version	Avg	Precision	Recall	F1
V1	Macro	0.5404	0.6376	0.4985
V1	Weighted	0.6818	0.4666	0.4466
V2	Macro	0.7104	0.8620	**0.7537**
V2	Weighted	0.8113	0.7416	**0.7457**

Table 3: Macro and weighted average metrics for the environment classifier, V1 vs. V2.

Class	V1 F1	V2 F1	Prec.	Rec.	Support
animal	0.635	**0.846**	0.769	0.941	6584
crowd	0.574	**0.658**	0.498	0.969	4413
indoor	0.537	**0.756**	0.939	0.633	36714
other_ambient	0.642	**0.800**	0.690	0.951	3134
outdoor_nature	0.382	**0.645**	0.495	0.926	10895
vehicle_ground	0.307	**0.725**	0.888	0.613	28300
vehicle_water	0.419	**0.755**	0.638	0.925	5771
water	0.493	**0.844**	0.767	0.938	8084

Table 4: Per-class metrics for the V2 environment classifier with V1 F1 for comparison. Every class improves in V2.

Every class improves in V2. The most dramatic gains occur in *vehicle_ground* (+0.418) and *water* (+0.351), the two classes most disadvantaged by V1's class imbalance. The primary remaining confusion is between *indoor* and *other_ambient*, which share overlapping background noise characteristics, and between *vehicle_ground* and *outdoor_nature*, which both occur in outdoor contexts.

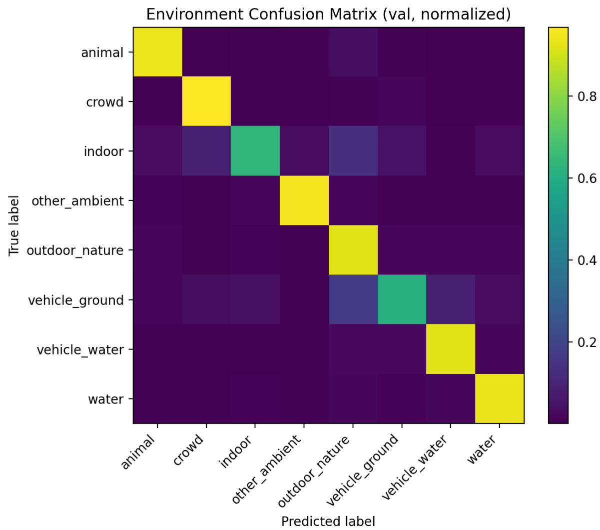

Figure 2: Normalized confusion matrix for the V2 8-class environment classifier. The V2 diagonal is notably cleaner than V1, with near-zero off-diagonal mass for most class pairs.

6.2 Event Classification Results

Table 5 reports overall metrics for the event detector. V2 improves macro-F1 from 0.39 to 0.55 despite tripling the label space from 62 to 208 classes — a meaningful result given that more classes generally increases task difficulty.

Version	Avg	Precision	Recall	F1
V1	Macro	0.33	0.62	0.39
V2	Macro	—	—	**0.550**
V2	Micro	0.584	0.723	**0.646**
V2	mAP		0.640	

Table 5: Overall event detection metrics, V1 vs. V2. V2 uses 208 classes vs. V1's 62.

Class	F1	Prec.	Rec.	Support
Wind/Woodwind Instr.	0.912	0.920	0.904	2653
Brass instrument	0.896	0.859	0.936	1035
Bowed string instr.	0.893	0.873	0.913	2064
Percussion	0.867	0.922	0.818	3977
Thunderstorm	0.875	0.804	0.958	601

Table 6: Top 5 event classes by F1-score in V2.

The pattern mirrors V1 in structure: acoustically distinctive classes with sufficient training examples perform well, while rare and acoustically overlapping classes remain difficult. The high recall but low precision on worst-case classes indicates the model identifies candidate events but generates false positives for similar-sounding classes.

Class	F1	Prec.	Rec.	Support
Tick	0.192	0.109	0.807	109
Truck	0.213	0.123	0.792	154
Fill (liquid)	0.224	0.128	0.926	148
Chuckle	0.225	0.129	0.883	111
Rattle	0.249	0.170	0.487	300

Table 7: Bottom 5 event classes by F1-score in V2.

6.3 Speech Classification Results

The speech classifier is new in V2 and has no V1 equivalent. Table 8 presents per-class and overall metrics.

Class	N	Prec.	Rec.	F1	Acc.
no_speech	447	0.321	0.978	0.483	0.978
single_speaker	25,718	0.999	0.966	0.982	0.966
multi_speaker	16,778	0.968	0.963	0.966	0.963
Overall accuracy			**96.5%**		
Macro F1			**0.810**		

Table 8: Per-class speech classifier metrics. The speech model is the strongest performer in the V2 pipeline.

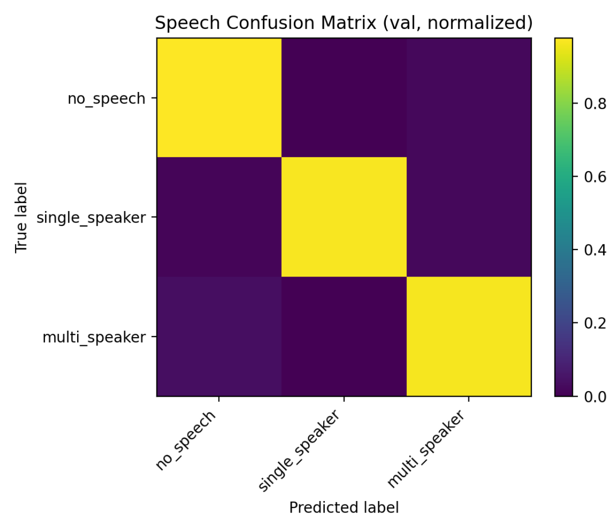

Figure 3: Normalized confusion matrix for the speech classifier. Single- and multi-speaker classes are cleanly separated; *no_speech* shows high recall but low precision due to class imbalance.

The *single_speaker* and *multi_speaker* classes are cleanly separated (F1 of 0.982 and 0.966 respectively). The primary weakness is *no_speech* precision (0.321), caused by severe class imbalance: only

447 *no_speech* clips were available versus 42K+ speech clips. The model compensates with high *no_speech* recall (0.978), meaning it rarely misses true silent segments, but frequently misclassifies ambiguous audio as speech.

Two additional limitations affect the speech classifier in live deployment. First, the model was trained on clean read speech (Common Voice) and multi-speaker home recordings (CHiME-6), leaving a gap for naturalistic conversational speech that falls between these extremes. Second, brief or intermittent speaker overlap can blur the *single_speaker*/*multi_speaker* boundary in real time, where the classification window may capture a transition rather than a stable condition.

6.4 V1 vs. V2 Summary

Model	V1 Macro-F1	V2 Macro-F1	Change
Environment (8 classes)	0.499	0.754	+0.255
Events (62→208 classes)	0.390	0.550	+0.160
Speech (new)	–	0.810	–

Table 9: V1 vs. V2 macro-F1 comparison across all tasks.

6.5 Per-Class Performance

Per-class metrics reveal patterns consistent across all three models. For the environment classifier:

- **Clear, high-SNR classes perform best.** Labels such as *animal* and *water* show high F1-scores because they contain distinct sound signatures.
- **Acoustically similar scenes cause confusion.** Classes like *indoor* and *other_ambient* overlap heavily in background noise patterns, leading to lower precision.
- **Rare categories underperform relative to frequent ones.** Despite SE attention improving all classes, *outdoor_nature* remains the weakest at F1 = 0.645.

For the event classifier:

- **Common events with distinct timbres generalize well.** Instruments such as brass, bowed strings, and woodwinds achieve F1 above 0.89 due to abundant training examples and clear spectral cues.
- **Fine-grained or co-occurring labels remain difficult.** Classes like *Tick*, *Truck*, and *Fill (with liquid)* suffer from low support and acoustic similarity to neighbouring categories.
- **Long-tailed categories struggle despite focal loss.** Focal loss reduces but does not eliminate the imbalance penalty for the rarest classes.

6.6 Discussion

The results highlight several practical challenges that shape performance across all three models:

- **Class imbalance remains a persistent issue.** Despite focal loss, weighted sampling, and SE attention, the rarest event classes still lack sufficient examples to learn stable representations.
- **Label noise affects the environment classifier.** Because AudioSet clips originate from YouTube, many samples contain incorrect labels, low-quality audio, or mismatched timestamps, blurring boundaries between categories such as *indoor* and *other_ambient*.
- **Lightweight CNNs have limited expressive capacity.** The models are intentionally compact for real-time use, which reduces their ability to differentiate fine-grained events or model long-term temporal structure.
- **Some confusion comes from genuine acoustic overlap.** Certain categories are simply hard to separate — *wind* vs. *ocean waves*, or the family of bird-related labels — even for a human listener.

Despite these limitations, all three classifiers are sufficiently stable for real-time use and produce coherent outputs when combined with the live microphone pipeline.

7 Supplemental: Real-Time Demo and Inference Pipeline

In addition to offline evaluation, Surround Sound V2 includes a real-time demonstration that classifies live microphone audio using all three models and produces live ASR transcription when speech is detected. Although not part of the formal grading criteria, the demo provides a practical illustration of how the full pipeline behaves in real scenarios.

7.1 Audio Streaming and Feature Extraction

The demo captures audio between 1 and 15 seconds and processes it through the same preprocessing pipeline described in Section 4. Each window is:

- recorded in real time at 16 kHz mono,
- normalized and padded to a fixed duration,
- converted into a log–mel spectrogram using the unified feature extractor.

Because the models assume 10-second inputs, the streaming version center-crops longer recordings to match the expected input length.

7.2 Three-Head Inference

Once a spectrogram window is computed, it is passed simultaneously to all three classifiers:

- The **environment head** produces an 8-class softmax distribution, estimating the most likely background scene.
- The **event head** outputs the top-5 sigmoid probabilities above a tuned threshold across 208 labels, generating a compact list of active events.
- The **speech head** produces a 3-class softmax distribution (*no_speech*, *single_speaker*, *multi_speaker*).

When the speech head predicts *single_speaker* or *multi_speaker* above a confidence threshold, the audio segment is additionally passed to Whisper large-v3 [12] via `faster-whisper` for full ASR transcription. Thresholds for both the event and speech heads are tuned on the validation set to balance recall and precision.

7.3 Natural-Language Scene Summarization

To present results in a human-friendly format, the demo optionally uses GPT-4.1 Mini to convert raw classifier outputs into a short textual summary. The summarizer receives:

- the top environment prediction and confidence,
- the subset of event labels above threshold,
- the speech class prediction, and
- the Whisper transcript, if available.

The LLM then generates a brief sentence such as: *"You appear to be indoors in a crowded space. A single speaker is talking about meeting schedules."* This summarization layer is an interpretability aid and does not influence any model output.

7.4 User Interface and Visualization

A Streamlit [1] interface displays the live waveform, log–mel spectrogram, all three classifier outputs with confidence values, the Whisper transcript when active, and the GPT-generated scene summary. Users can start or stop live recording, inspect prediction trends, and observe how small acoustic changes influence the models. Although the interface is lightweight, it provides valuable insight into the behavior of all three classifiers under real-time constraints.

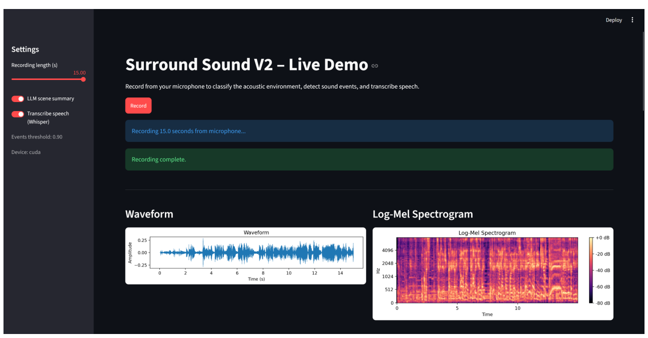

Figure 4: Surround Sound V2 live demo: recording interface, waveform, and log–mel spectrogram for a 15-second microphone capture.

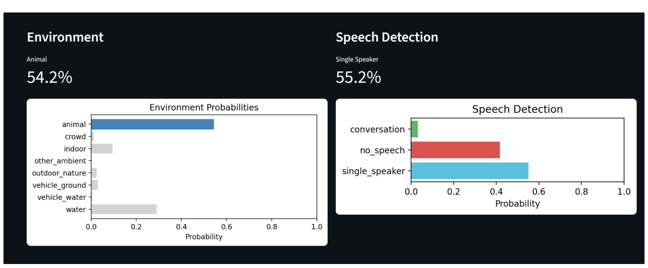

Figure 5: Environment classifier (Animal, 54.2%) and speech classifier (Single Speaker, 55.2%) outputs for the same recording.

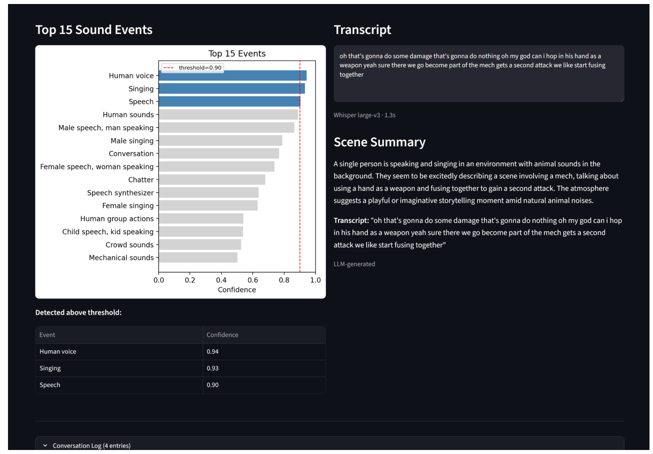

Figure 6: Event detector, Whisper large-v3 transcript, and GPT-4.1 Mini scene summary. Human voice (0.94), Singing (0.93), and Speech (0.90) are detected above threshold; the transcript captures dialogue from a D&D session over video game background music.

7.5 Pipeline Overview

Figure 7 summarizes the architecture of the real-time system, from audio capture through three-head classification, Whisper transcription, and GPT-4.1 Mini scene summarization.

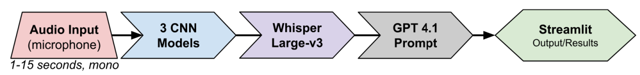

Figure 7: Overview of the V2 real-time inference pipeline.

In summary, the demo highlights the strengths and limitations of all three classifiers when deployed in a realistic streaming context. Despite occasional noisiness in event outputs and the *no_speech* precision weakness discussed in Section 6, the models successfully capture broad scene characteristics, identify salient events, detect speech conditions, and transcribe spoken content — demonstrating the feasibility of low-latency multi-task auditory scene understanding.

8 Conclusion and Future Work

This project introduced Surround Sound V2, a three-head audio classification system capable of identifying background environments, fine-grained sound events, and speech conditions from both offline audio and real-time microphone input. Building on the two-head V1 architecture, V2 adds a dedicated speech classifier, integrates Whisper large-v3 for live ASR transcription, expands the sound event taxonomy from 62 to 208 classes, and applies a suite of modern deep learning improvements across all models. Using a unified preprocessing pipeline and lightweight CNN architectures with SE attention and residual connections, the system achieves substantial performance gains over V1: environment macro-F1 rises from 0.50 to 0.75, event macro-F1 rises from 0.39 to 0.55 despite tripling the label space, and the new speech classifier achieves 96.5% accuracy.

The experimental results highlight several persistent challenges, including dataset imbalance, label noise, and acoustic overlap between similar classes. The *no_speech* precision weakness in the speech classifier is a direct consequence of data scarcity rather than architecture, and the long-tail event classes remain difficult despite focal loss and weighted sampling. The real-time demo illustrates how all three classifiers behave in dynamic acoustic environments and demonstrates the practical feasibility of low-latency multi-task auditory scene understanding.

Future work will focus on several directions:

- **Addressing no_speech data scarcity.** Collecting additional acoustically diverse non-speech segments would directly improve speech classifier precision without architectural changes.
- **Exploring transformer-based architectures.** Models such as the Audio Spectrogram Transformer (AST) or pre-trained audio encoders could capture longer temporal dependencies and richer spectral patterns while remaining deployable in real time.
- **Hierarchical event classification.** Exploiting the AudioSet ontology structure to organize related labels into parent–child relationships could reduce confusion between fine-grained co-occurring classes and improve rare-class performance.
- **Speaker diarization.** Extending the speech head from *multi_speaker* detection to full diarization would identify who is speaking when, substantially enriching the scene description.
- **Training models specifically for streaming audio.** Teaching the models to operate on shorter or shifting windows would improve stability at the boundaries of live recording segments.
- **Tighter multi-task integration.** Unifying the three heads into a shared-encoder, multi-task model could allow each task to benefit from the representations learned by the others, reducing redundant computation and potentially improving all three tasks simultaneously.

Overall, Surround Sound V2 provides a flexible and extensible framework for multi-level auditory scene understanding. With further refinement, the system has the potential to support robust, context-aware audio interpretation — combining environmental awareness, acoustic event detection, and spoken language understanding into a single cohesive pipeline.

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